Cobalamin (vitamin B_{12}) is a water-soluble, essential micronutrient crucial for many biological processes such as cellular metabolism, the production of red blood cells, and the function of several cellular enzymes within enzyme-catalyzed reactions. The chemical structure of vitamin B_{12} is among the most complex of all the vitamins, consisting of: (1) a central cobalt(III) atom covalently bonded to an R group, (2) a corrin ring with various methyl and amide substituents surrounding the cobalt(III) atom, (3) a 5,6-dimethylbenzimidazole ligand below the corrin ring, and (4) a pentose sugar with a phosphate linkage attached to one of the amide substituents of the corrin ring. These features of the chemical structure of the vitamin B_{12} molecule are illustrated in Figure 1.

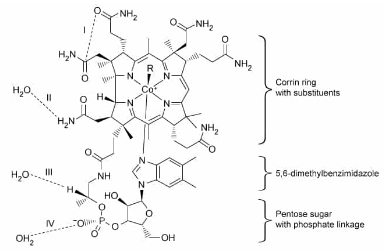


Figure 1 Chemical structure of cobalamin (vitamin B_{12}) in an aqueous solution (Note: Dashed lines I, II, III, and IV indicate noncovalent interactions. The length of the Co–N bond with 5,6-dimethylbenzimidazole is exaggerated to better visualize of all parts of the structure in the diagram.)

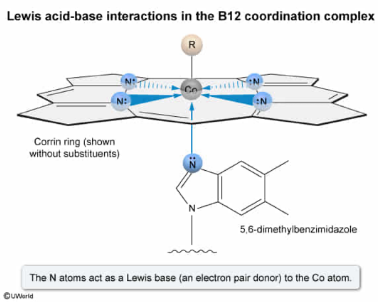
The identity of the R group influences its function and determines which of the four common forms (vitamers) the molecule takes. Synthetic forms are produced as cyanocobalamin (R = –CN) whereas biologically active forms exist as hydroxocobalamin (R = –OH), methylcobalamin (R = $–CH_3$), or 5-deoxyadenosylcobalamin (R = 5-deoxyadenosyl).

Which of the following statements correctly describes part of the formation of the cobalt complex in the chemical structure of vitamin B_{12}?

- A. The 5,6-dimethylbenzimidazole group acts as an Arrhenius acid toward the Co atom. [3%]
- B. The corrin ring acts as a Brønsted-Lowry acid toward the Co atom. [8%]
- C. The Co atom acts as a Lewis base toward the 5,6-dimethylbenzimidazole group. [20%]
- D. The N atoms of the corrin ring act as a Lewis base toward the Co atom. [66%]

Correct
66% answered correctly

Explanation:



Coordination complexes consist of a central metal ion surrounded by one or more ions or molecules called **ligands**, which are bound to the metal center by a coordinate covalent bond. The ligands act as a Lewis base (ie, an electron pair donor), and the metal center acts as a **Lewis acid** (ie, an electron pair acceptor). Ligand structures that can form more than one coordinate bond by donating more than one pair of electrons have more than one atom that can function as a Lewis base.

Therefore, in the structure of vitamin B_{12}, the four nitrogen atoms of the corrin ring each act as a Lewis base toward the Co atom by donating two electrons to form a coordinate covalent bond (ie, a total of four pairs of electrons are donated to form four coordinate bonds).

(Choices A and B) Neither the corrin ring nor the 5,6-dimethylbenzimidazole group act as an Arrhenius acid or Brønsted-Lowry acid (ie, H^+ donors) because no hydrogen ion is transferred to the Co atom. The corrin ring and 5,6-dimethylbenzimidazole group both act as a *Lewis base* (ie, electron pair donor) toward the Co atom.

(Choice C) As stated in the passage, the central Co atom is cobalt(III) (ie, the atom has a +3 oxidation state and is electron deficient). As such, the Co atom cannot donate electrons (ie, cannot act as a Lewis base) to the N atom of the 5,6-dimethylbenzimidazole group. Instead, the Co atom acts as a Lewis *acid*.

Educational objective:
Within coordination complexes, a ligand acts as a Lewis base and the metal center acts as a Lewis acid. Ligand structures that can form more than one coordinate bond by donating more than one pair of electrons have more than one site that can function as a Lewis base.

Time spent:0 sec
Subject:
General Chemistry

QID:403318
System:
Interactions of Chemical Substances

Cobalamin (vitamin B_{12}) is a water-soluble, essential micronutrient crucial for many biological processes such as cellular metabolism, the production of red blood cells, and the function of several cellular enzymes within enzyme-catalyzed reactions. The chemical structure of vitamin B_{12} is among the most complex of all the vitamins, consisting of: (1) a central cobalt(III) atom covalently bonded to an R group, (2) a corrin ring with various methyl and amide substituents surrounding the cobalt(III) atom, (3) a 5,6-dimethylbenzimidazole ligand below the corrin ring, and (4) a pentose sugar with a phosphate linkage attached to one of the amide substituents of the corrin ring. These features of the chemical structure of the vitamin B_{12} molecule are illustrated in Figure 1.

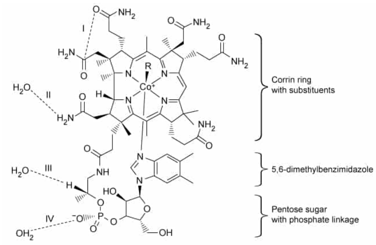


Figure 1 Chemical structure of cobalamin (vitamin B_{12}) in an aqueous solution (Note: Dashed lines I, II, III, and IV indicate noncovalent interactions. The length of the Co–N bond with 5,6-dimethylbenzimidazole is exaggerated to better visualize of all parts of the structure in the diagram.)

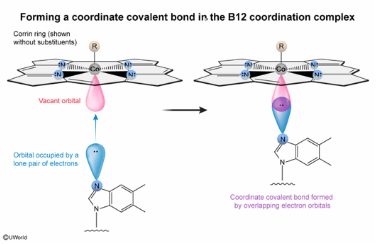
The identity of the R group influences its function and determines which of the four common forms (vitamers) the molecule takes. Synthetic forms are produced as cyanocobalamin (R = –CN) whereas biologically active forms exist as hydroxocobalamin (R = –OH), methylcobalamin (R = $-CH_3$), or 5-deoxyadenosylcobalamin (R = 5-deoxyadenosyl).

Based on the information given in the passage, which of the following statements is true about the Co–N bond formed between the cobalt atom and the nitrogen atom of the 5,6-dimethylbenzimidazole group in the vitamin B_{12} complex?

- A. The Co–N bond forms by a Co^{3+} ion donating two electrons to the N atom. [4%]
- B. The Co–N bond forms by the N atom donating one electron to the Co atom. [16%]
- ✔ C. The Co–N bond is a coordinate covalent bond formed by overlapping electron orbitals. [56%]
- D. The Co–N bond is an ionic bond between a metal and a nonmetal. [23%]

Correct
55% answered correctly

Explanation:



A coordinate covalent bond forms by a Lewis acid-base interaction between an electron-deficient metal cation (ie, the Lewis acid) and an electron-rich atom (ie, the Lewis base) of a ligand. In effect, a **lone pair of electrons** occupying an orbital of an **electron-rich atom** in the ligand plugs into a vacant orbital of the **electron-deficient metal** cation to share the electrons and establish the coordinate covalent bond. As such, the bond forms by the overlap of the two orbitals, and the ligand donates both electrons to form the coordinate bond.

Therefore, in the structure of vitamin B_{12}, the Co–N bond between the cobalt atom and the nitrogen atom of the 5,6-dimethylbenzimidazole group forms when the orbitals from the Co and N atoms overlap as the electron-rich N atom donates its lone pair of electrons to the electron-deficient Co atom.

(Choice A) A Co^{3+} ion is electron-deficient and cannot donate electrons to the N atom of the 5,6-dimethylbenzimidazole group. Instead, the electron-deficient Co atom accepts two electrons from the electron-rich N atom.

(Choice B) The Co–N bond forms by the N atom donating *two electrons* (ie, its nonbonding electron pair) to the Co atom.

(Choice D) Although the Co–N bond is between a metal and a nonmetal, the bond is a *coordinate covalent bond* formed by sharing electrons. No negative ion is formed in the Lewis acid-base coordination.

Educational objective:
A coordinate covalent bond forms by a Lewis acid-base interaction in which a lone pair of electrons from an electron-rich atom in a ligand is shared with an electron-deficient metal cation via the overlap of atomic orbitals without forming additional ions.

Time spent:0 sec
Subject:
General Chemistry

QID:403319
System:
Interactions of Chemical Substances

Cobalamin (vitamin B_{12}) is a water-soluble, essential micronutrient crucial for many biological processes such as cellular metabolism, the production of red blood cells, and the function of several cellular enzymes within enzyme-catalyzed reactions. The chemical structure of vitamin B_{12} is among the most complex of all the vitamins, consisting of: (1) a central cobalt(III) atom covalently bonded to an R group, (2) a corrin ring with various methyl and amide substituents surrounding the cobalt(III) atom, (3) a 5,6-dimethylbenzimidazole ligand below the corrin ring, and (4) a pentose sugar with a phosphate linkage attached to one of the amide substituents of the corrin ring. These features of the chemical structure of the vitamin B_{12} molecule are illustrated in Figure 1.

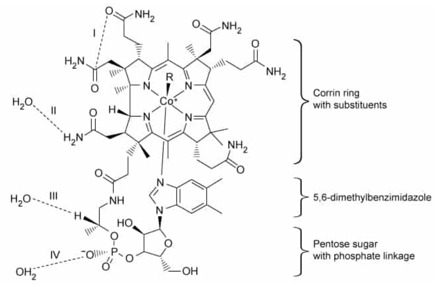


Figure 1 Chemical structure of cobalamin (vitamin B_{12}) in an aqueous solution (Note: Dashed lines I, II, III, and IV indicate noncovalent interactions. The length of the Co–N bond with 5,6-dimethylbenzimidazole is exaggerated to better visualize of all parts of the structure in the diagram.)

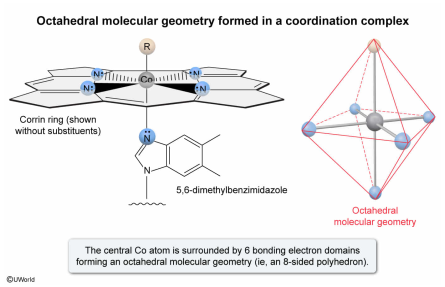
The identity of the R group influences its function and determines which of the four common forms (vitamers) the molecule takes. Synthetic forms are produced as cyanocobalamin (R = –CN) whereas biologically active forms exist as hydroxocobalamin (R = –OH), methylcobalamin (R = $–CH_3$), or 5-deoxyadenosylcobalamin (R = 5-deoxyadenosyl).

The molecular geometry of the central cobalt complex in the center of vitamin B_{12} is closest to:

- A. square planar. [3%]
- B. octahedral. [72%]
- C. trigonal bipyramidal. [14%]
- D. square pyramidal. [10%]

Correct
71% answered correctly

Explanation:



According to valence-shell electron-pair repulsion (VSEPR) theory, the electron pairs around a central atom in a molecule will adopt an electron geometry (ie, a spatial arrangement) that maximizes the separation between the electron domains and minimizes the repulsion between the electrons. Using VSEPR theory, the molecular geometry (ie, the shape) of a molecule can be predicted from its Lewis structure by evaluating the orientations that the bonded atoms will adopt due to the number of bonding and nonbonding electron domains present around the central atom.

In the structure of vitamin B_{12}, the central Co atom is surrounded by six electron domains (ie, 6 bonds and 0 nonbonding electron pairs). As such, VSEPR theory predicts an octahedral electron geometry, and because all the electron domains around the Co atom are bonding electrons, the molecular geometry is also predicted to be **octahedral**.

(Choice A) A square planar molecular geometry would result if the Co atom had 4 bonds and 2 nonbonding electron pairs (eg, if Co was only bonded to the nitrogen atoms of the corrin ring, and the R group and 5,6-dimethylbenzimidazole group were replaced by nonbonding lone pairs).

(Choice C) A trigonal bipyramidal molecular geometry would result if the Co atom had 5 electron domains consisting of 5 bonds and 0 nonbonding electron pairs.

(Choice D) A square pyramidal molecular geometry would result if the Co atom had 5 bonds and 1 nonbonding electron pair (eg, if the R group or the 5,6-dimethylbenzimidazole group were replaced by a nonbonding pair of electrons).

Educational objective:
Valence-shell electron-pair repulsion (VSEPR) theory holds that the electron domains around a central atom will adopt an electron geometry that maximizes the separation and minimizes the repulsion between the electrons. The molecular geometry (shape) of a molecule depends on the electron geometry and number of nonbonding electron pairs of the central atom.

Time spent:0 sec
Subject:
General Chemistry

QID:403320
System:
Interactions of Chemical Substances

Cobalamin (vitamin B_{12}) is a water-soluble, essential micronutrient crucial for many biological processes such as cellular metabolism, the production of red blood cells, and the function of several cellular enzymes within enzyme-catalyzed reactions. The chemical structure of vitamin B_{12} is among the most complex of all the vitamins, consisting of: (1) a central cobalt(III) atom covalently bonded to an R group, (2) a corrin ring with various methyl and amide substituents surrounding the cobalt(III) atom, (3) a 5,6-dimethylbenzimidazole ligand below the corrin ring, and (4) a pentose sugar with a phosphate linkage attached to one of the amide substituents of the corrin ring. These features of the chemical structure of the vitamin B_{12} molecule are illustrated in Figure 1.

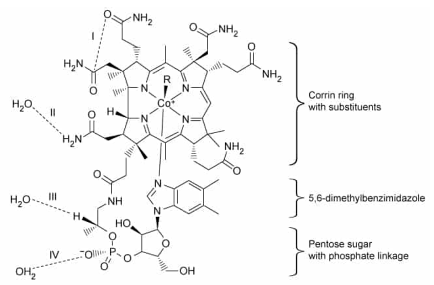
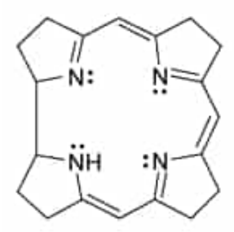


Figure 1 Chemical structure of cobalamin (vitamin B_{12}) in an aqueous solution (Note: Dashed lines I, II, III, and IV indicate noncovalent interactions. The length of the Co–N bond with 5,6-dimethylbenzimidazole is exaggerated to better visualize of all parts of the structure in the diagram.)

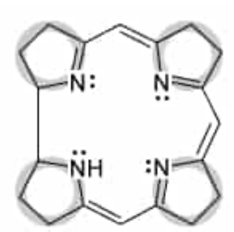
The identity of the R group influences its function and determines which of the four common forms (vitamers) the molecule takes. Synthetic forms are produced as cyanocobalamin (R = –CN) whereas biologically active forms exist as hydroxocobalamin (R = –OH), methylcobalamin (R = $–CH_3$), or 5-deoxyadenosylcobalamin (R = 5-deoxyadenosyl).

The structure of an unbound corrin ring without substituents is shown below.

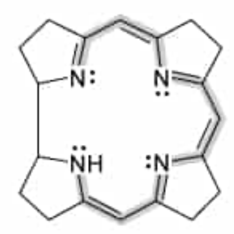


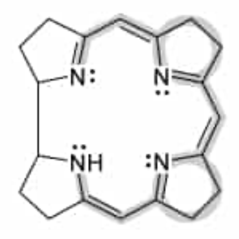
Which of the following structural diagrams highlights only the atoms that are a part of a conjugated π system within the corrin ring?

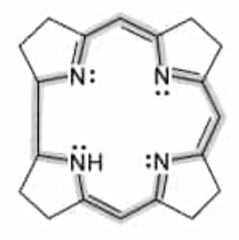
○ A.


[2%]

✔ B.

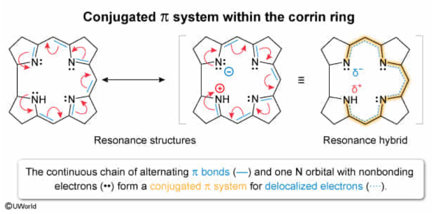

[81%]

○ C.  [6%]

○ D.  [9%]

Correct
81% answered correctly

Explanation:



Some molecules cannot be fully represented using a single Lewis structure because some of the electrons can adopt more than one valid bonding configuration; this effect is known as resonance. In resonance, the arrangement of the atoms is unchanged but some π bonds and nonbonding electrons can "oscillate" from one atom to an adjacent atom in ways that do not violate the octet rule (apart from a valid exception). In effect, resonance causes the electrons to be delocalized (ie, "spread out") among the participating atoms.

For compounds that undergo resonance, a single Lewis structure only represents the bond configuration at a moment in time (ie, a resonance contributor). Because the electrons are in different places at different times, the true molecular structure is a resonance hybrid (ie, a weighted average) of all resonance contributors.

Conjugation refers to a system of three or more *p* orbitals linked together through resonance that function as a single network for delocalized electrons. Nonbonding (lone-pair) electrons or electron vacancies (ie, positive charges) at correct alternate positions may also participate in a conjugated sequence. Atoms that are conjugated provide a continuous pathway along which the electrons can be delocalized.

A conjugated system in a molecular structure is often observed as a chain of alternating single and multiple (ie, double or triple) bonds that can participate together in resonance. In the corrin ring, a continuous chain of alternating double and single bonds extends most of the way around the ring's interior. The nonbonding electrons on one of the N atoms located at the end of the chain are also part of the alternating pattern. Therefore, this pathway (including the nonbonding electron pair) is conjugated as shown by the diagram in Choice B.

(Choices A, C, and D) Each of these diagrams highlights portions of the structure that cannot participate in resonance with the primary conjugated chain.

Educational objective:
Conjugation refers to a system of three or more *p* orbitals linked together through resonance that function as a single network for delocalized electrons. A conjugated system is often observed in a molecular structure as a chain of alternating single and multiple bonds that can participate together in resonance.

Time spent:0 sec
Subject:
General Chemistry

QID:403321
System:
Interactions of Chemical Substances

Cobalamin (vitamin B_{12}) is a water-soluble, essential micronutrient crucial for many biological processes such as cellular metabolism, the production of red blood cells, and the function of several cellular enzymes within enzyme-catalyzed reactions. The chemical structure of vitamin B_{12} is among the most complex of all the vitamins, consisting of: (1) a central cobalt(III) atom covalently bonded to an R group, (2) a corrin ring with various methyl and amide substituents surrounding the cobalt(III) atom, (3) a 5,6-dimethylbenzimidazole ligand below the corrin ring, and (4) a pentose sugar with a phosphate linkage attached to one of the amide substituents of the corrin ring. These features of the chemical structure of the vitamin B_{12} molecule are illustrated in Figure 1.

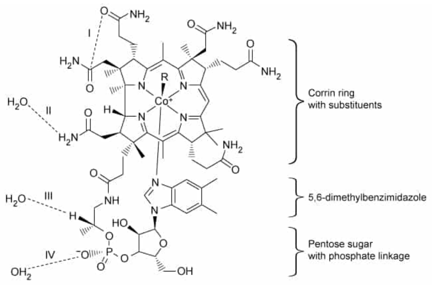


Figure 1 Chemical structure of cobalamin (vitamin B_{12}) in an aqueous solution (Note: Dashed lines I, II, III, and IV indicate noncovalent interactions. The length of the Co–N bond with 5,6-dimethylbenzimidazole is exaggerated to better visualize of all parts of the structure in the diagram.)

The identity of the R group influences its function and determines which of the four common forms (vitamers) the molecule takes. Synthetic forms are produced as cyanocobalamin (R = –CN) whereas biologically active forms exist as hydroxocobalamin (R = –OH), methylcobalamin (R = $–CH_3$), or 5-deoxyadenosylcobalamin (R = 5-deoxyadenosyl).

The magnitude of a dipole moment μ across a polar bond can be approximated by the equation:

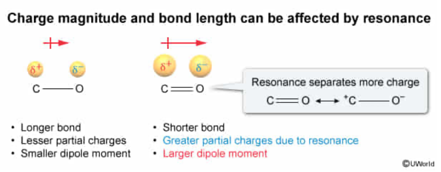
$$\mu = qr$$

where q is the magnitude of the partial charge separated by a distance r. In the structure of vitamin B_{12}, the average length of a C=O bond (121 pm) in the amide groups is shorter than the length of the C–O bond (143 pm) in the pentose sugar, but the C=O bond has a larger dipole moment than the C–O bond. Which of the following is the most likely explanation for this observation?

- A. The difference in electronegativity between the atoms in C–O and C=O bonds is not the same. [6%]
- B. The longer C–O bond length decreases the force of attraction between the charges. [41%]
- C. The influence from resonance acts to decrease the C=O bond length. [7%]
- ✔ D. The effect of resonance increases the amount of charge separated across the C=O bond. [44%]

Correct
45% answered correctly

Explanation:



The magnitude of the dipole moment μ of a polar bond is equal to the product of the magnitude of the partial charge q and the charge separation distance r (ie, the chemical bond length):

$$\mu = qr$$

In general, atoms with a greater difference in electronegativity have more separated charge and form stronger dipoles than other bonds of comparable length. Relative bond lengths (taken as the sum of the atomic radii) can often be assessed qualitatively based on periodic trends in the atomic radii of the atoms and on the bond order (ie, higher bond order indicates a shorter bond).

The higher bond order rightly predicts that the C=O bond is shorter than the C–O bond. The electronegativity difference between the atoms is the same in both bonds, and this suggests the same amount of separated charge across both bonds; however, the electronegativity difference does not account for the effects of resonance in the C=O bond. The π bond in C=O is capable of momentary delocalization to the O atom via resonance.

$$C{=}O \leftrightarrow {}^{+}C{-}O^{-}$$

Therefore, the effect of resonance increases the average amount of charge separated across the carbon-oxygen bond. Although the C=O bond is shorter, the larger magnitude of separated charge gives the C=O bond a larger dipole moment.

(Choice A) Each bond consists of the same two elements (ie, C and O); therefore, the electronegativity difference between the atoms must be the *same* in both bonds.

(Choice B) Although the force of attraction between opposite charges does decrease as the separation distance increases (ie, Coulomb's law), the *dipole moment* (as defined by the given equation $\mu = qr$) *increases* as charge separation distance increases. As such, bond length alone cannot explain the observation.

(Choice C) The resonance hybrid of a bond is the weighted average of all resonance contributors. Single bonds are longer than double bonds. Because the C=O bond can momentarily assume the $^{+}C{-}O^{-}$ form by resonance, the influence of resonance acts to *increase* the C=O bond length.

Educational objective:
The magnitude of the dipole moment of a polar bond is the product of the magnitude of the partial charge and its separation distance. These variables of the dipole moment can be influenced by resonance.

Time spent:0 sec
Subject:
General Chemistry

QID:403322
System:
Interactions of Chemical Substances

Cobalamin (vitamin B_{12}) is a water-soluble, essential micronutrient crucial for many biological processes such as cellular metabolism, the production of red blood cells, and the function of several cellular enzymes within enzyme-catalyzed reactions. The chemical structure of vitamin B_{12} is among the most complex of all the vitamins, consisting of: (1) a central cobalt(III) atom covalently bonded to an R group, (2) a corrin ring with various methyl and amide substituents surrounding the cobalt(III) atom, (3) a 5,6-dimethylbenzimidazole ligand below the corrin ring, and (4) a pentose sugar with a phosphate linkage attached to one of the amide substituents of the corrin ring. These features of the chemical structure of the vitamin B_{12} molecule are illustrated in Figure 1.

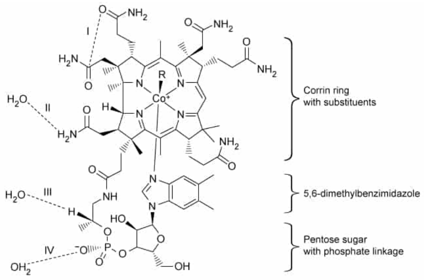


Figure 1 Chemical structure of cobalamin (vitamin B_{12}) in an aqueous solution (Note: Dashed lines I, II, III, and IV indicate noncovalent interactions. The length of the Co–N bond with 5,6-dimethylbenzimidazole is exaggerated to better visualize of all parts of the structure in the diagram.)

The identity of the R group influences its function and determines which of the four common forms (vitamers) the molecule takes. Synthetic forms are produced as cyanocobalamin (R = –CN) whereas biologically active forms exist as hydroxocobalamin (R = –OH), methylcobalamin (R = $–CH_3$), or 5-deoxyadenosylcobalamin (R = 5-deoxyadenosyl).

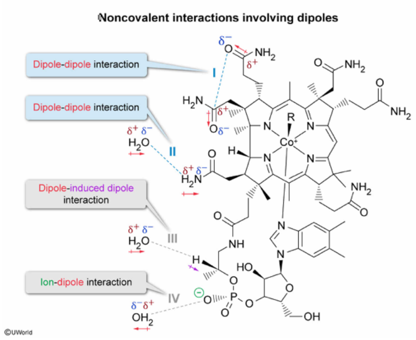
Which of the labeled interactions in Figure 1 are dipole-dipole interactions between two permanent dipoles?

(Note: Excludes induced dipoles and ions.)

- A. I and II only [31%]
- B. II and III only [20%]
- C. I, II, and III only [25%]
- D. I, II, III, and IV [23%]

Correct
30% answered correctly

Explanation:



A **dipole** consists of opposite electric charges separated across a distance (eg, a chemical bond). In molecules, permanent dipoles form across some covalent bonds due to unequal sharing of electrons between two atoms caused by large differences in electronegativity. In a polar bond, the region with greater electron density gains a partial negative charge (δ^-), and the region with less electron density acquires a partial positive charge (δ^+).

Many intramolecular and intermolecular forces (ie, noncovalent interactions between atoms and molecules) involve the partial charges of bond dipoles.

- Dipole-dipole interactions occur when the opposite partial charges of the permanent dipoles of two polar bonds form a mutual attraction, as depicted by **Interaction I**.
- Hydrogen bonding is a special type of dipole-dipole interaction that occurs between very strong permanent dipoles involving H atoms bonded to highly electronegative N, O, or F atoms, as depicted by **Interaction II**.
- Dipole–induced dipole interactions happen when the permanent dipole from the polar bond distorts the electron cloud of a neighboring nonpolar bond (eg, a C–H bond) and induces a weak temporary dipole in the nonpolar bond. The dipole and the induced dipole thus form a momentary attraction, as depicted by **Interaction III**.
- Ion-dipole interactions involve the full charge of an ion being attracted to the opposite partial charge of a permanent dipole in a polar bond, as depicted by **Interaction IV**.

Therefore, **Interactions I and II** are dipole-dipole interactions that involve only polar bonds with permanent dipole moments.

(Choices B, C, and D) Although each of the interactions involves at least one permanent dipole, Interaction III also involves a temporary, *induced dipole* in a nonpolar C–H bond, and Interaction IV involves an ionic charge.

Educational objective:
Dipole-dipole interactions occur between the partial charges of two permanent dipoles. Hydrogen bonding occurs between very strong dipoles involving H atoms bonded to N, O, or F atoms. Dipole–induced dipole interactions occur when a permanent dipole induces a weak temporary dipole in a nonpolar bond or atom. Ion-dipole interactions occur between the partial charge of a permanent dipole and the full charge of an ion.

Time spent:0 sec
Subject:
General Chemistry

QID:403323
System:
Interactions of Chemical Substances

Cobalamin (vitamin B_{12}) is a water-soluble, essential micronutrient crucial for many biological processes such as cellular metabolism, the production of red blood cells, and the function of several cellular enzymes within enzyme-catalyzed reactions. The chemical structure of vitamin B_{12} is among the most complex of all the vitamins, consisting of: (1) a central cobalt(III) atom covalently bonded to an R group, (2) a corrin ring with various methyl and amide substituents surrounding the cobalt(III) atom, (3) a 5,6-dimethylbenzimidazole ligand below the corrin ring, and (4) a pentose sugar with a phosphate linkage attached to one of the amide substituents of the corrin ring. These features of the chemical structure of the vitamin B_{12} molecule are illustrated in Figure 1.

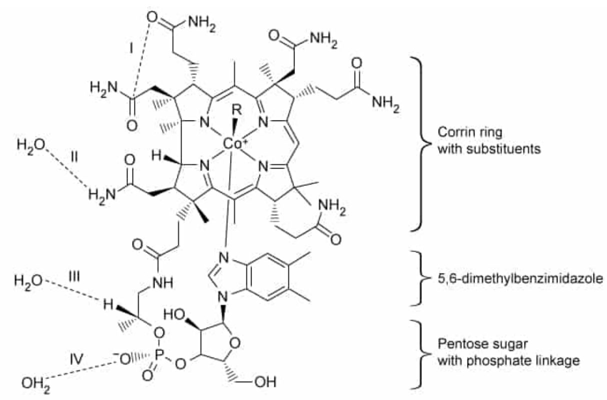


Figure 1 Chemical structure of cobalamin (vitamin B_{12}) in an aqueous solution (Note: Dashed lines I, II, III, and IV indicate noncovalent interactions. The length of the Co–N bond with 5,6-dimethylbenzimidazole is exaggerated to better visualize of all parts of the structure in the diagram.)

The identity of the R group influences its function and determines which of the four common forms (vitamers) the molecule takes. Synthetic forms are produced as cyanocobalamin (R = –CN) whereas biologically active forms exist as hydroxocobalamin (R = –OH), methylcobalamin (R = $–CH_3$), or 5-deoxyadenosylcobalamin (R = 5-deoxyadenosyl).

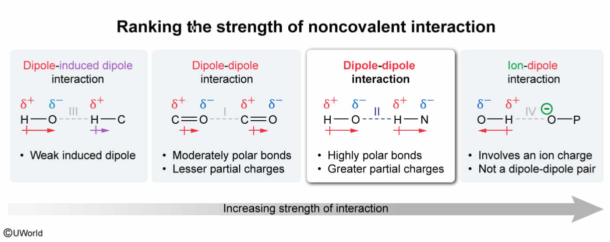
Which of the labeled interactions in Figure 1 is the strongest dipole-dipole interaction between two permanent dipoles? (Note: Excludes induced dipoles and ions.)

- A. I [27%]
- ✔ B. II [44%]
- C. III [5%]
- D. IV [22%]

Correct

43% answered correctly

Explanation:



Dipole-dipole interactions occur between molecules when the partial negative charge (δ^-) of one chemical bond dipole forms a mutual attraction to the partial positive charge (δ^+) of another chemical bond dipole. These dipolar interactions occur between permanent dipoles that form due to unequal sharing of electrons between two atoms with large differences in electronegativity.

Among the given interactions, only Interactions I and II are interactions between two permanent dipoles, and Interaction II is the strongest because the difference in electronegativity between the atoms of the O–H and N–H bonds is much greater the difference in the C=O bond. The large difference in electronegativity allows O–H and N–H bonds to participate is hydrogen bonding (a special type of dipole-dipole interaction that is exceptionally strong).

(Choice A) The electronegativity difference in a C=O bond is much smaller than in an O–H or N–H, which makes Interaction I weaker than Interaction II.

(Choice C) Interaction III is a dipole–induced dipole interaction and is the *weakest* of the four given interactions because an induced dipole is caused by a small distortion in the electron cloud of an atom, making the induced dipole very weak.

(Choice D) Interaction IV is the strongest of the four interactions because it involves more charge, but it is not a *dipole-dipole interaction* between two permanent dipoles, as specified in the question. Interaction IV is an ion-dipole interaction caused by the attraction of the partial charge of a permanent dipole to the full charge of an ion.

Educational objective:
The strength of dipole-dipole interactions is proportional to the magnitudes of the bond dipole moments, which can often be inferred by the difference in electronegativity of the bonded atoms.

Time spent:0 sec
Subject:
General Chemistry

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System:
Interactions of Chemical Substances